Near-strict 1-spike-per-cycle ceiling: 99.45% of (cell, cycle) pairs

nb046 · 4 June 2026

Abstract

nb041's slope $p \approx 0.20$ was interpreted as *"each E cell joins a cycle with ≈ 20% probability"*. That reading assumes the per-cycle spike count is bounded by 1 — an E cell either participates in this cycle or not. This notebook measures that directly: walking through every gamma cycle on the test set and counting how many spikes each E cell actually emits, on all 18 trained checkpoints in nb041's τ_{GABA} sweep.

Methods

For each of nb041's 18 trained cells (6 τ_{GABA} × 3 seeds):

1. Run inference on the MNIST test set; capture per-trial (T, B, N_E) and (T, B, N_I) spike tensors.
2. Detect I-burst times per trial: smooth the population I rate with a 1-ms Gaussian, run scipy peak detection with min-distance set to half the cell's own $1/f_\gamma$.
3. Cycle boundaries are the midpoints between consecutive I-burst peaks (first cycle starts at $t = 0$, last ends at trial end).
4. For each (cell, cycle, trial), count the number of E spikes within the cycle window.
5. Bucket counts globally into $\{0, 1, 2, \geq 3\}$ and aggregate by τ_{GABA}.

The cycle anchor is the I-burst — this is the right anchor because the cycle is operationally defined as *"the time between one inhibitory blanket and the next"*, not as the time between E bursts (which can be silent on a given cycle).

Results

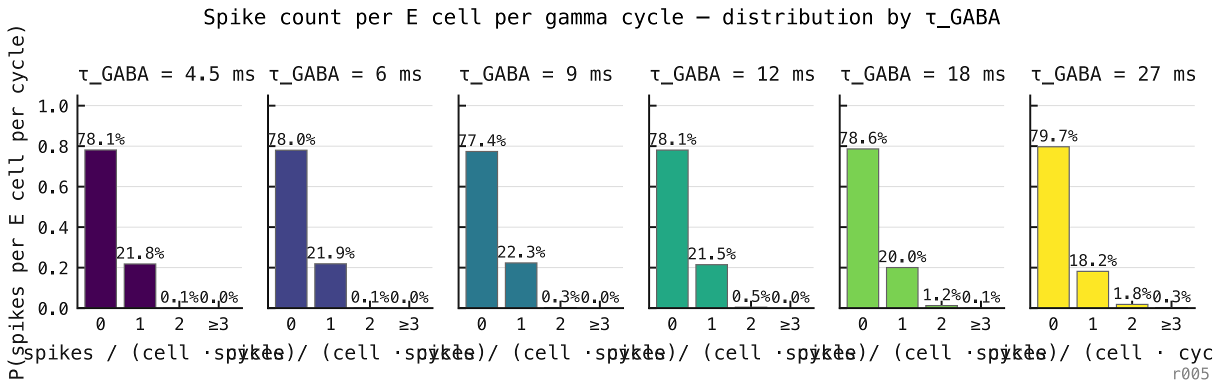

Figure 136: Distribution of E spike count per gamma cycle per cell, by τ_{GABA}, three seeds aggregated. Across **48 million (cell, cycle) pairs**, the architecture is overwhelmingly bimodal: each cell either emits zero spikes in a given cycle (≈ 77% of the time) or exactly one (≈ 22% of the time). Two-or-more events occur in 0.55% of cycles; three-or-more in 0.04%.